Pre-Analysis Plan

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HYPOTHESIS TEST FOR H1

Hypothesis: Having daughters leads to stronger support for gender equality, as measured by attitudes.

1) Variables

- Dependent variable: plh0301
    - This variable captures agreement with the statement: “Women should rather care about family than career.”
    - It is reverse-coded so that higher values indicate greater disagreement with traditional norms, i.e., greater support for women's career rights.
    - Values of -1, -5, and -8 are treated as missing or invalid and will be excluded.
- Key independent variable: d_daughter
    - A dummy variable coded as 1 if the respondent has at least one daughter, 0 otherwise.
- Control variables:
    - sex (1 = male, 2 = female)
    - gebjahr (year of birth, continuous)
    - pgfamstd (marital status, included as a categorical variable using i.pgfamstd)
    - pgbilzeit (years of education, continuous)

2) Inclusion/Exclusion Criteria

- Year restriction: The analysis is restricted to 2012, which is the only wave containing both plh0301 and d_daughter.

- Exclusion of missing/invalid observations:
    - plh0301: exclude if equal to -8, -5, or -1
    - pgfamstd: exclude if equal to -3 or -1
    - pgbilzeit: exclude if equal to -2 or -1

3) Exact Analysis

- We estimate the following linear OLS regression:

reg plh0301 d_daughter sex gebjahr i.pgfamstd pgbilzeit if (plh0301 != -8) & (plh0301 != -5) & (plh0301 != -1) & (pgfamstd != -3) & (pgfamstd != -1) & (pgbilzeit != -2) & (pgbilzeit != -1), r

4) Interpretation of Results

- We focus on the coefficient of d_daughter, which represents the effect of having a daughter on support for gender equality.
- A positive coefficient indicates that parents with daughters express greater support for women's career rights. However, it is not statistically significant.

HYPOTHESIS TEST FOR H2

Hypothesis: Having daughters leads to more left-leaning political preferences.

1) Variables

- Dependent variable: plh0004
    - This variable captures self-placement on a left–right political scale, where respondents locate themselves on a unidimensional ideological continuum.
    - A lower score corresponds to more left-leaning views, while a higher score indicates more right-leaning preferences.
    - Responses coded as -1, -5, and -8 are considered invalid or missing and will be excluded.
- Key independent variable: d_daughter
    - A binary variable equal to 1 if the respondent has at least one daughter, 0 otherwise.
- Control variables:
    - sex (1 = male, 2 = female)
    - gebjahr (year of birth, continuous)
    - pgfamstd (marital status, categorical, entered as i.pgfamstd)
    - pgbilzeit (years of education, continuous)
    - syear (survey year, included as fixed effects using i.syear) to control for time trends across waves.

2) Inclusion/Exclusion Criteria

- Year restriction: The analysis is limited to the 2009, 2014, and 2019 waves, which are the only years that contain both plh0004 and d_daughter.
- Exclusion of invalid/missing observations:
    - plh0004: exclude if equal to -8, -5, or -1
    - pgfamstd: exclude if equal to -3 or -1

- pgbilzeit: exclude if equal to -2 or -1

3) Exact Analysis

- We estimate the following linear OLS regression:

reg plh0004 d_daughter sex gebjahr i.pgfamstd pgbilzeit i.syear if (plh0004 != -8) & (plh0004 != -5) & (plh0004 != -1) & (pgfamstd != -3) & (pgfamstd != -1) & (pgbilzeit != -2) & (pgbilzeit != -1), r

4) Interpretation of Results

- The main statistic of interest is the coefficient on d_daughter.
- A negative and (weakly) statistically significant coefficient ($p < 0.10$) indicates that having daughters is associated with more left-leaning political preferences, consistent with the hypothesis.

HYPOTHESIS TEST FOR H3

Hypothesis: Having daughters leads to more gender equality in the household, as measured by behaviors or time-use outcomes.

1) Variables

- Dependent variable: pli0043_h
    - This variable captures the number of housework hours performed by the respondent on a typical weekday.
    - A higher value reflects more time spent on domestic labor, used here as a behavioral proxy for gender roles in the household.
    - Values coded as -1, -5, or -8 are treated as invalid and excluded from analysis.
- Key independent variables:
    - d_daughter: Dummy variable equal to 1 if the respondent has at least one daughter, 0 otherwise.
    - sex: Respondent's gender (1 = male, 2 = female)
    - An interaction term d_daughter × sex is included to test whether the effect of having daughters differs by gender.
- Control variables:
    - gebjahr (year of birth, continuous)
    - pgfamstd (marital status, categorical, entered as i.pgfamstd)
    - pgbilzeit (years of education, continuous)
    - syear (included as fixed effects using i.syear) to absorb time trends in housework norms

2) Inclusion/Exclusion Criteria

- Year restriction: All years where pli0043_h and d_daughter are jointly available will be included.
- Exclusion of invalid/missing observations:

- pli0043_h: exclude if equal to -8, -5, or -1
- pgfamstd: exclude if equal to -3 or -1
- pgbilzeit: exclude if equal to -2 or -1

3) Exact Analysis

- The model is estimated using a fixed effects panel regression with an interaction term:

keep

if (pgfamstd != -3) & (pgfamstd != -1)

xtreg pli0043_h d_daughter##sex gebjahr i.pgfamstd pgbilzeit i.syear if (pli0043_h != -8) & (pli0043_h != -5) & (pli0043_h != -1) & (pgbilzeit != -2) & (pgbilzeit != -1), fe r

restore

4) Interpretation of Results

- The main coefficient of interest is the interaction term: d_daughter#sex.
- A negative and statistically significant coefficient on the interaction term suggests that women with daughters increase their housework hours less than men, who already bear a larger share of domestic work. This finding is in line with the hypothesis implies that having daughters promotes more gender-equal division of household labor, primarily through behavior change among male respondents.

HYPOTHESIS TEST FOR H4

Hypothesis: Having daughters leads to a larger increase in support for gender equality among men than among women, as measured by attitudes.

1) Variables

- Dependent variable: plh0301
    - This variable captures agreement with the statement: "Women should rather care about family than career."
    - It is reverse-coded, such that higher values reflect stronger disagreement with traditional norms, and thus greater support for women's career rights.
    - Responses coded as -1, -5, and -8 will be excluded.
- Key independent variables:
    - d_daughter: Dummy variable equal to 1 if the respondent has at least one daughter, 0 otherwise.
    - sex: Dummy variable indicating respondent gender (1 = male, 2 = female)
    - The interaction term d_daughter × sex captures whether the effect of having a daughter differs by gender.
- Control variables:
    - gebjahr (year of birth, continuous)
    - pgfamstd (marital status, categorical, included as i.pgfamstd)
    - pgbilzeit (years of education, continuous)

2) Inclusion/Exclusion Criteria

- Year restriction: The analysis is restricted to the 2012 wave, which is the only year that includes both plh0301 and d_daughter.
- Exclusion of invalid/missing responses:
    - plh0301: exclude if equal to -8, -5, or -1
    - pgfamstd: exclude if equal to -3 or -1
    - pgbilzeit: exclude if equal to -2 or -1

3) Exact Analysis

- We estimate the following OLS regression model with an interaction term:

reg plh0301 i.d_daughter##i.sex gebjahr i.pgfamstd pgbilzeit if (plh0301 != -8) & (plh0301 != -5) & (plh0301 != -1) & (pgfamstd != -3) & (pgfamstd != -1) & (pgbilzeit != -2) & (pgbilzeit != -1), r

4) Interpretation of Results

- The main coefficient of interest is the interaction term between d_daughter and sex (i.e., 1.d_daughter#2.sex).
- A negative and statistically significant interaction coefficient indicates that the effect of having a daughter on gender-equal attitudes is weaker for women than for men, which is consistent with the hypothesis. However, the result is not statistically significant.